

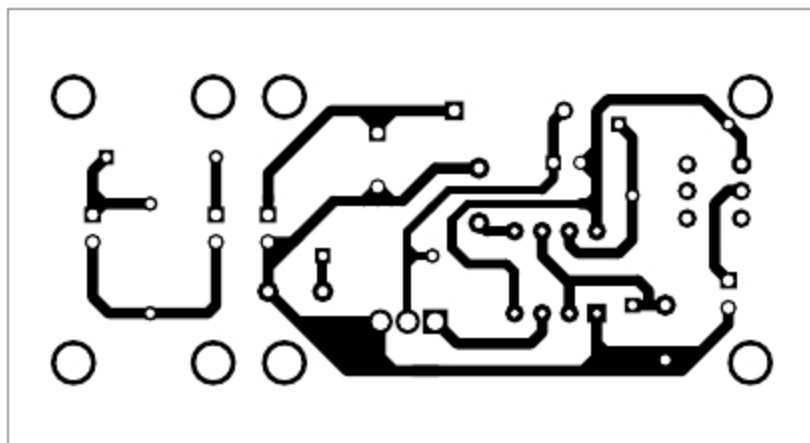


Fig. 3: Actual-size PCB layout of high-voltage generator

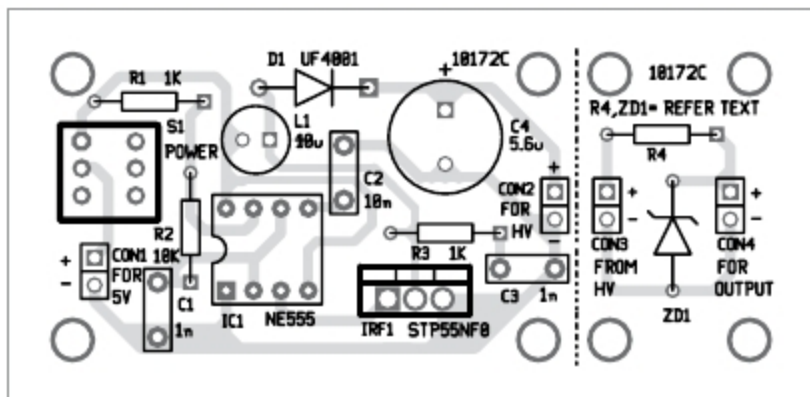


Fig. 4: Components layout for the PCB

(Ireqd) at 27V (Vreqd). For this, use a Zener diode (ZD1) rated at 27V, and calculate the value of series/ballast

in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, connect 5V across CON1.

resistor R4 using the formula given below:

$$R4 = \frac{V_{in} - V_{reqd}}{I_{reqd}} = \frac{50V - 27V}{1mA}$$

(in this example)

Values of R4 and ZD1 are based on your requirement.

This is a basic design idea for electronics hobbyists. You can modify the circuit by changing the operating frequency of NE555 IC, and/or altering component values like inductor, MOSFET, rectifier and filter capacitor, to get desired results.

Construction and testing

An actual-size PCB layout for the high-voltage generator is shown

PARTS LIST

Semiconductors:

- IC1 - NE555 timer IC
- D1 - UF4001 diode
- IRF1 - STP55NF0 MOSFET
- ZD1 - Refer text

Resistors (all 1/4-watt, ±5% carbon):

- R1, R3 - 1-kilo-ohm
- R2 - 10-kilo-ohm
- R4 - Refer text

Capacitors:

- C1, C3 - 1nF ceramic disk
- C2 - 10nF ceramic disk
- C4 - 5.6μF, 250V electrolytic

Miscellaneous:

- CON1-CON4 - 2-pin terminal connector
- L1 - 10μH inductor with ferrite core
- S1 - On/off switch
- 5V DC regulated power supply

Caution. This circuit may not be as dangerous as AC mains power, but it can still shock you depending on your body resistance. So, take necessary precautions while working on this circuit. **EFY**



T.K. Hareendran is founder and promoter of TechNode Protolabz

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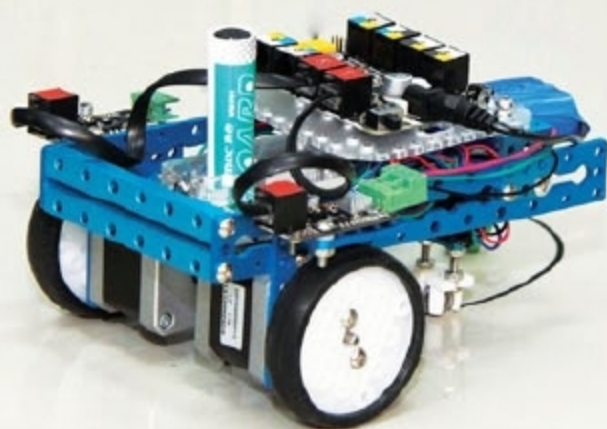
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